Worksheet#5a

Espadon, Gerona, Bernasol

2024-11-23

```r
# Load necessary libraries
library(rvest)
library(httr)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

```r
library(polite)
library(stringr)
library(ggplot2)
```

```r
# Define target URL
url <- 'https://www.imdb.com/chart/toptv/'

# Create a polite session
session <- bow(url, user_agent = "Educational")

session
```

```
## <polite session> https://www.imdb.com/chart/toptv/
##     User-agent: Educational
##     robots.txt: 35 rules are defined for 3 bots
##    Crawl delay: 5 sec
##   The path is scrapable for this user-agent
```

```r
#Extracting the ranks and titles
title_list <- read_html(url) %>%
html_nodes('.ipc-title__text') %>%
html_text()

title_list
```

```
##  [1] "IMDb Charts"
##  [2] "Top 250 TV Shows"
##  [3] "1. Breaking Bad"
##  [4] "2. Planet Earth II"
##  [5] "3. Planet Earth"
```

```
##  [6] "4. Band of Brothers"
##  [7] "5. Chernobyl"
##  [8] "6. The Wire"
##  [9] "7. Avatar: The Last Airbender"
## [10] "8. Blue Planet II"
## [11] "9. The Sopranos"
## [12] "10. Cosmos: A Spacetime Odyssey"
## [13] "11. Cosmos"
## [14] "12. Our Planet"
## [15] "13. Game of Thrones"
## [16] "14. Bluey"
## [17] "15. The World at War"
## [18] "16. Fullmetal Alchemist: Brotherhood"
## [19] "17. Rick and Morty"
## [20] "18. Life"
## [21] "19. The Last Dance"
## [22] "20. The Twilight Zone"
## [23] "21. The Vietnam War"
## [24] "22. Sherlock"
## [25] "23. Attack on Titan"
## [26] "24. Batman: The Animated Series"
## [27] "25. Arcane"
## [28] "Recently viewed"
```

```r
#Cleaning extracted text
title_list_sub <- as.data.frame(title_list[3:27], stringsAsFactors = FALSE)
colnames(title_list_sub) <- "ranks"

split_df <- strsplit(as.character(title_list_sub$ranks), "\\.", fixed = FALSE)
split_df <- data.frame(do.call(rbind, split_df), stringsAsFactors = FALSE)

colnames(split_df) <- c("rank", "title")
split_df <- split_df %>%
select(rank, title)

split_df$title <- trimws(split_df$title)

rank_title <- split_df
rank_title
```

```
##    rank                            title
## 1     1                     Breaking Bad
## 2     2                  Planet Earth II
## 3     3                     Planet Earth
## 4     4                 Band of Brothers
## 5     5                        Chernobyl
## 6     6                         The Wire
## 7     7       Avatar: The Last Airbender
## 8     8                    Blue Planet II
## 9     9                     The Sopranos
## 10   10      Cosmos: A Spacetime Odyssey
## 11   11                           Cosmos
## 12   12                       Our Planet
## 13   13                  Game of Thrones
## 14   14                            Bluey
```

```
## 15   15                 The World at War
## 16   16 Fullmetal Alchemist: Brotherhood
## 17   17                   Rick and Morty
## 18   18                             Life
## 19   19                   The Last Dance
## 20   20                The Twilight Zone
## 21   21                  The Vietnam War
## 22   22                         Sherlock
## 23   23                  Attack on Titan
## 24   24      Batman: The Animated Series
## 25   25                           Arcane
```

```
# Scrape Ratings
#Extracting tv rating, the number of people who voted, the number of episodes, and the year it was rele
rating_ls <- read_html(url) %>%
html_nodes('.ipc-rating-star--rating') %>%
html_text()

rating_ls
```

```
##  [1] "9.5" "9.5" "9.4" "9.4" "9.3" "9.3" "9.3" "9.3" "9.2" "9.2" "9.3" "9.2"
## [13] "9.2" "9.3" "9.2" "9.1" "9.1" "9.1" "9.0" "9.0" "9.1" "9.1" "9.1" "9.0"
## [25] "9.0"
```

```
# Scrape Vote Counts
voter_ls <- read_html(url) %>%
html_nodes('.ipc-rating-star--voteCount') %>%
html_text()

clean_votes <- gsub('[()]', '', voter_ls)
# Check if vote counts were extracted correctly
print(voter_ls)
```

```
##  [1] " (2.2M)" " (163K)" " (224K)" " (547K)" " (910K)" " (392K)" " (391K)"
##  [8] " (49K)"  " (500K)" " (132K)" " (46K)"  " (54K)"  " (2.4M)" " (34K)"
## [15] " (32K)"  " (210K)" " (629K)" " (44K)"  " (160K)" " (97K)"  " (30K)"
## [22] " (1M)"   " (565K)" " (123K)" " (332K)"
```

```
#extracted the number of episodes
eps_ls <- read_html(url) %>%
html_nodes('span.sc-300a8231-7.eaXxft.cli-title-metadata-item:nth-of-type(2)') %>%
html_text()
clean_eps <- gsub('[eps]', '', eps_ls)

num_eps <- as.numeric(clean_eps)

print(num_eps)
```

```
##  [1]  62   6  11  10   5  60  62   7  86  13  13  12  74 194  26  68  78  11  10
## [20] 156  10  15  98  85  18
```

```
#note to self, use gsub() to remove constant strings appearing in the dataset.
```

```
#extracted the year released
years <- read_html(url) %>%
html_nodes('span.sc-300a8231-7.eaXxft.cli-title-metadata-item:nth-of-type(1)') %>%
html_text()
```

```r
years
```

```
##  [1] "2008-2013" "2016"      "2006"      "2001"      "2019"      "2002-2008"
##  [7] "2005-2008" "2017"      "1999-2007" "2014"      "1980"      "2019-2023"
## [13] "2011-2019" "2018- "    "1973-1974" "2009-2010" "2013- "    "2009"
## [19] "2020"      "1959-1964" "2017"      "2010-2017" "2013-2023" "1992-1995"
## [25] "2021-2024"
```

```r
top_tv_shows <- data.frame(
Title = rank_title[,2],
Rating = rating_ls,
Voters = clean_votes,
Episodes = num_eps,
Year = years)


top_tv_shows
```

```
##                                Title Rating Voters Episodes      Year
## 1                       Breaking Bad    9.5   2.2M       62 2008-2013
## 2                    Planet Earth II    9.5   163K        6      2016
## 3                       Planet Earth    9.4   224K       11      2006
## 4                   Band of Brothers    9.4   547K       10      2001
## 5                          Chernobyl    9.3   910K        5      2019
## 6                           The Wire    9.3   392K       60 2002-2008
## 7         Avatar: The Last Airbender    9.3   391K       62 2005-2008
## 8                     Blue Planet II    9.3    49K        7      2017
## 9                       The Sopranos    9.2   500K       86 1999-2007
## 10      Cosmos: A Spacetime Odyssey    9.2   132K       13      2014
## 11                            Cosmos    9.3    46K       13      1980
## 12                        Our Planet    9.2    54K       12 2019-2023
## 13                  Game of Thrones    9.2   2.4M       74 2011-2019
## 14                             Bluey    9.3    34K      194    2018-
## 15                  The World at War    9.2    32K       26 1973-1974
## 16 Fullmetal Alchemist: Brotherhood    9.1   210K       68 2009-2010
## 17                    Rick and Morty    9.1   629K       78    2013-
## 18                              Life    9.1    44K       11      2009
## 19                    The Last Dance    9.0   160K       10      2020
## 20                 The Twilight Zone    9.0    97K      156 1959-1964
## 21                   The Vietnam War    9.1    30K       10      2017
## 22                          Sherlock    9.1     1M       15 2010-2017
## 23                   Attack on Titan    9.1   565K       98 2013-2023
## 24      Batman: The Animated Series    9.0   123K       85 1992-1995
## 25                            Arcane    9.0   332K       18 2021-2024
```

```r
home_link <- 'https://www.imdb.com/chart/toptv/'
main_page <- read_html(home_link)
```

```r
links <- main_page %>%
html_nodes("a.ipc-title-link-wrapper") %>%
html_attr("href")

links
```

```
##  [1] "/title/tt0903747/?ref_=chttvtp_t_1"
```

```
##  [2] "/title/tt5491994/?ref_=chttvtp_t_2"
##  [3] "/title/tt0795176/?ref_=chttvtp_t_3"
##  [4] "/title/tt0185906/?ref_=chttvtp_t_4"
##  [5] "/title/tt7366338/?ref_=chttvtp_t_5"
##  [6] "/title/tt0306414/?ref_=chttvtp_t_6"
##  [7] "/title/tt0417299/?ref_=chttvtp_t_7"
##  [8] "/title/tt6769208/?ref_=chttvtp_t_8"
##  [9] "/title/tt0141842/?ref_=chttvtp_t_9"
## [10] "/title/tt2395695/?ref_=chttvtp_t_10"
## [11] "/title/tt0081846/?ref_=chttvtp_t_11"
## [12] "/title/tt9253866/?ref_=chttvtp_t_12"
## [13] "/title/tt0944947/?ref_=chttvtp_t_13"
## [14] "/title/tt7678620/?ref_=chttvtp_t_14"
## [15] "/title/tt0071075/?ref_=chttvtp_t_15"
## [16] "/title/tt1355642/?ref_=chttvtp_t_16"
## [17] "/title/tt2861424/?ref_=chttvtp_t_17"
## [18] "/title/tt1533395/?ref_=chttvtp_t_18"
## [19] "/title/tt8420184/?ref_=chttvtp_t_19"
## [20] "/title/tt0052520/?ref_=chttvtp_t_20"
## [21] "/title/tt1877514/?ref_=chttvtp_t_21"
## [22] "/title/tt1475582/?ref_=chttvtp_t_22"
## [23] "/title/tt2560140/?ref_=chttvtp_t_23"
## [24] "/title/tt0103359/?ref_=chttvtp_t_24"
## [25] "/title/tt11126994/?ref_=chttvtp_t_25"
```

```r
# Loop to get link of each show's page
show_data <- lapply(links, function(link) {
complete_link <- paste0("https://imdb.com", link)

#loop to get the link for user review page
usrv_link <- read_html(complete_link)
usrv_link_page <- usrv_link %>%
html_nodes('a.isReview') %>%
html_attr("href")

#loop to extract critic reviews
critic <- usrv_link %>%
html_nodes("span.score") %>%
html_text()
critic_df <- data.frame(Critic_Reviews = critic[2], stringsAsFactors = FALSE)

#loop to extract pop rating
pop_rating <- usrv_link %>%
html_nodes('[data-testid="hero-rating-bar__popularity__score"]') %>%
html_text()

#loop to get user reviews of each shows
usrv <- read_html(paste0("https://imdb.com", usrv_link_page[1]))
usrv_count <- usrv %>%
html_nodes('[data-testid="tturv-total-reviews"]') %>%
html_text()

return(data.frame(Show_Link = complete_link, User_Reviews = usrv_count, Critic = critic_df, Popularity_
})
```

```r
show_url_df <- do.call(rbind, show_data)
print(show_url_df)
```

```
##                                              Show_Link  User_Reviews
## 1    https://imdb.com/title/tt0903747/?ref_=chttvtp_t_1 5,108 reviews
## 2    https://imdb.com/title/tt0903747/?ref_=chttvtp_t_1 5,108 reviews
## 3    https://imdb.com/title/tt5491994/?ref_=chttvtp_t_2   158 reviews
## 4    https://imdb.com/title/tt5491994/?ref_=chttvtp_t_2   158 reviews
## 5    https://imdb.com/title/tt0795176/?ref_=chttvtp_t_3   111 reviews
## 6    https://imdb.com/title/tt0795176/?ref_=chttvtp_t_3   111 reviews
## 7    https://imdb.com/title/tt0185906/?ref_=chttvtp_t_4 1,058 reviews
## 8    https://imdb.com/title/tt0185906/?ref_=chttvtp_t_4 1,058 reviews
## 9    https://imdb.com/title/tt7366338/?ref_=chttvtp_t_5 3,534 reviews
## 10   https://imdb.com/title/tt7366338/?ref_=chttvtp_t_5 3,534 reviews
## 11   https://imdb.com/title/tt0306414/?ref_=chttvtp_t_6   787 reviews
## 12   https://imdb.com/title/tt0306414/?ref_=chttvtp_t_6   787 reviews
## 13   https://imdb.com/title/tt0417299/?ref_=chttvtp_t_7 1,003 reviews
## 14   https://imdb.com/title/tt0417299/?ref_=chttvtp_t_7 1,003 reviews
## 15   https://imdb.com/title/tt6769208/?ref_=chttvtp_t_8    53 reviews
## 16   https://imdb.com/title/tt6769208/?ref_=chttvtp_t_8    53 reviews
## 17   https://imdb.com/title/tt0141842/?ref_=chttvtp_t_9   966 reviews
## 18   https://imdb.com/title/tt0141842/?ref_=chttvtp_t_9   966 reviews
## 19  https://imdb.com/title/tt2395695/?ref_=chttvtp_t_10   205 reviews
## 20  https://imdb.com/title/tt2395695/?ref_=chttvtp_t_10   205 reviews
## 21  https://imdb.com/title/tt0081846/?ref_=chttvtp_t_11    80 reviews
## 22  https://imdb.com/title/tt0081846/?ref_=chttvtp_t_11    80 reviews
## 23  https://imdb.com/title/tt9253866/?ref_=chttvtp_t_12   245 reviews
## 24  https://imdb.com/title/tt9253866/?ref_=chttvtp_t_12   245 reviews
## 25  https://imdb.com/title/tt0944947/?ref_=chttvtp_t_13 5,907 reviews
## 26  https://imdb.com/title/tt0944947/?ref_=chttvtp_t_13 5,907 reviews
## 27  https://imdb.com/title/tt7678620/?ref_=chttvtp_t_14   369 reviews
## 28  https://imdb.com/title/tt7678620/?ref_=chttvtp_t_14   369 reviews
## 29  https://imdb.com/title/tt0071075/?ref_=chttvtp_t_15   126 reviews
## 30  https://imdb.com/title/tt0071075/?ref_=chttvtp_t_15   126 reviews
## 31  https://imdb.com/title/tt1355642/?ref_=chttvtp_t_16   468 reviews
## 32  https://imdb.com/title/tt1355642/?ref_=chttvtp_t_16   468 reviews
## 33  https://imdb.com/title/tt2861424/?ref_=chttvtp_t_17   910 reviews
## 34  https://imdb.com/title/tt2861424/?ref_=chttvtp_t_17   910 reviews
## 35  https://imdb.com/title/tt1533395/?ref_=chttvtp_t_18    12 reviews
## 36  https://imdb.com/title/tt1533395/?ref_=chttvtp_t_18    12 reviews
## 37  https://imdb.com/title/tt8420184/?ref_=chttvtp_t_19   542 reviews
## 38  https://imdb.com/title/tt8420184/?ref_=chttvtp_t_19   542 reviews
## 39  https://imdb.com/title/tt0052520/?ref_=chttvtp_t_20   214 reviews
## 40  https://imdb.com/title/tt0052520/?ref_=chttvtp_t_20   214 reviews
## 41  https://imdb.com/title/tt1877514/?ref_=chttvtp_t_21   175 reviews
## 42  https://imdb.com/title/tt1877514/?ref_=chttvtp_t_21   175 reviews
## 43  https://imdb.com/title/tt1475582/?ref_=chttvtp_t_22 1,098 reviews
## 44  https://imdb.com/title/tt1475582/?ref_=chttvtp_t_22 1,098 reviews
## 45  https://imdb.com/title/tt2560140/?ref_=chttvtp_t_23 2,370 reviews
## 46  https://imdb.com/title/tt2560140/?ref_=chttvtp_t_23 2,370 reviews
## 47  https://imdb.com/title/tt0103359/?ref_=chttvtp_t_24   219 reviews
## 48  https://imdb.com/title/tt0103359/?ref_=chttvtp_t_24   219 reviews
## 49 https://imdb.com/title/tt11126994/?ref_=chttvtp_t_25 2,217 reviews
## 50 https://imdb.com/title/tt11126994/?ref_=chttvtp_t_25 2,217 reviews
```

```
##    Critic_Reviews Popularity_Rating
## 1             175                20
## 2             175                20
## 3               6               999
## 4               6               999
## 5              10             1,778
## 6              10             1,778
## 7              34               153
## 8              34               153
## 9              88               146
## 10             88               146
## 11             77               104
## 12             77               104
## 13             57               357
## 14             57               357
## 15              9             4,122
## 16              9             4,122
## 17             93                31
## 18             93                31
## 19             12             1,571
## 20             12             1,571
## 21              8             3,645
## 22              8             3,645
## 23             15             2,401
## 24             15             2,401
## 25            368                16
## 26            368                16
## 27              4               373
## 28              4               373
## 29              5             2,532
## 30              5             2,532
## 31             16               474
## 32             16               474
## 33             94               125
## 34             94               125
## 35              9             3,057
## 36              9             3,057
## 37             28             1,403
## 38             28             1,403
## 39             85               337
## 40             85               337
## 41             13             1,739
## 42             13             1,739
## 43            121               160
## 44            121               160
## 45             64                50
## 46             64                50
## 47             25               463
## 48             25               463
## 49             59                 1
## 50             59                 1
```

```
shows <- cbind(top_tv_shows, show_url_df)
shows
```

```
##                                Title Rating Voters Episodes      Year
## 1                       Breaking Bad    9.5   2.2M       62 2008-2013
## 2                    Planet Earth II    9.5   163K        6      2016
## 3                       Planet Earth    9.4   224K       11      2006
## 4                   Band of Brothers    9.4   547K       10      2001
## 5                          Chernobyl    9.3   910K        5      2019
## 6                           The Wire    9.3   392K       60 2002-2008
## 7         Avatar: The Last Airbender    9.3   391K       62 2005-2008
## 8                     Blue Planet II    9.3    49K        7      2017
## 9                       The Sopranos    9.2   500K       86 1999-2007
## 10      Cosmos: A Spacetime Odyssey    9.2   132K       13      2014
## 11                            Cosmos    9.3    46K       13      1980
## 12                        Our Planet    9.2    54K       12 2019-2023
## 13                   Game of Thrones    9.2   2.4M       74 2011-2019
## 14                             Bluey    9.3    34K      194    2018-
## 15                  The World at War    9.2    32K       26 1973-1974
## 16 Fullmetal Alchemist: Brotherhood    9.1   210K       68 2009-2010
## 17                    Rick and Morty    9.1   629K       78    2013-
## 18                              Life    9.1    44K       11      2009
## 19                    The Last Dance    9.0   160K       10      2020
## 20                 The Twilight Zone    9.0    97K      156 1959-1964
## 21                   The Vietnam War    9.1    30K       10      2017
## 22                          Sherlock    9.1     1M       15 2010-2017
## 23                   Attack on Titan    9.1   565K       98 2013-2023
## 24      Batman: The Animated Series    9.0   123K       85 1992-1995
## 25                            Arcane    9.0   332K       18 2021-2024
## 26                       Breaking Bad    9.5   2.2M       62 2008-2013
## 27                    Planet Earth II    9.5   163K        6      2016
## 28                       Planet Earth    9.4   224K       11      2006
## 29                   Band of Brothers    9.4   547K       10      2001
## 30                          Chernobyl    9.3   910K        5      2019
## 31                           The Wire    9.3   392K       60 2002-2008
## 32         Avatar: The Last Airbender    9.3   391K       62 2005-2008
## 33                     Blue Planet II    9.3    49K        7      2017
## 34                       The Sopranos    9.2   500K       86 1999-2007
## 35      Cosmos: A Spacetime Odyssey    9.2   132K       13      2014
## 36                            Cosmos    9.3    46K       13      1980
## 37                        Our Planet    9.2    54K       12 2019-2023
## 38                   Game of Thrones    9.2   2.4M       74 2011-2019
## 39                             Bluey    9.3    34K      194    2018-
## 40                  The World at War    9.2    32K       26 1973-1974
## 41 Fullmetal Alchemist: Brotherhood    9.1   210K       68 2009-2010
## 42                    Rick and Morty    9.1   629K       78    2013-
## 43                              Life    9.1    44K       11      2009
## 44                    The Last Dance    9.0   160K       10      2020
## 45                 The Twilight Zone    9.0    97K      156 1959-1964
## 46                   The Vietnam War    9.1    30K       10      2017
## 47                          Sherlock    9.1     1M       15 2010-2017
## 48                   Attack on Titan    9.1   565K       98 2013-2023
## 49      Batman: The Animated Series    9.0   123K       85 1992-1995
## 50                            Arcane    9.0   332K       18 2021-2024
##                                                Show_Link  User_Reviews
## 1    https://imdb.com/title/tt0903747/?ref_=chttvtp_t_1 5,108 reviews
## 2    https://imdb.com/title/tt0903747/?ref_=chttvtp_t_1 5,108 reviews
```

```
## 3    https://imdb.com/title/tt5491994/?ref_=chttvtp_t_2   158 reviews
## 4    https://imdb.com/title/tt5491994/?ref_=chttvtp_t_2   158 reviews
## 5    https://imdb.com/title/tt0795176/?ref_=chttvtp_t_3   111 reviews
## 6    https://imdb.com/title/tt0795176/?ref_=chttvtp_t_3   111 reviews
## 7    https://imdb.com/title/tt0185906/?ref_=chttvtp_t_4 1,058 reviews
## 8    https://imdb.com/title/tt0185906/?ref_=chttvtp_t_4 1,058 reviews
## 9    https://imdb.com/title/tt7366338/?ref_=chttvtp_t_5 3,534 reviews
## 10   https://imdb.com/title/tt7366338/?ref_=chttvtp_t_5 3,534 reviews
## 11   https://imdb.com/title/tt0306414/?ref_=chttvtp_t_6   787 reviews
## 12   https://imdb.com/title/tt0306414/?ref_=chttvtp_t_6   787 reviews
## 13   https://imdb.com/title/tt0417299/?ref_=chttvtp_t_7 1,003 reviews
## 14   https://imdb.com/title/tt0417299/?ref_=chttvtp_t_7 1,003 reviews
## 15   https://imdb.com/title/tt6769208/?ref_=chttvtp_t_8    53 reviews
## 16   https://imdb.com/title/tt6769208/?ref_=chttvtp_t_8    53 reviews
## 17   https://imdb.com/title/tt0141842/?ref_=chttvtp_t_9   966 reviews
## 18   https://imdb.com/title/tt0141842/?ref_=chttvtp_t_9   966 reviews
## 19  https://imdb.com/title/tt2395695/?ref_=chttvtp_t_10   205 reviews
## 20  https://imdb.com/title/tt2395695/?ref_=chttvtp_t_10   205 reviews
## 21  https://imdb.com/title/tt0081846/?ref_=chttvtp_t_11    80 reviews
## 22  https://imdb.com/title/tt0081846/?ref_=chttvtp_t_11    80 reviews
## 23  https://imdb.com/title/tt9253866/?ref_=chttvtp_t_12   245 reviews
## 24  https://imdb.com/title/tt9253866/?ref_=chttvtp_t_12   245 reviews
## 25  https://imdb.com/title/tt0944947/?ref_=chttvtp_t_13 5,907 reviews
## 26  https://imdb.com/title/tt0944947/?ref_=chttvtp_t_13 5,907 reviews
## 27  https://imdb.com/title/tt7678620/?ref_=chttvtp_t_14   369 reviews
## 28  https://imdb.com/title/tt7678620/?ref_=chttvtp_t_14   369 reviews
## 29  https://imdb.com/title/tt0071075/?ref_=chttvtp_t_15   126 reviews
## 30  https://imdb.com/title/tt0071075/?ref_=chttvtp_t_15   126 reviews
## 31  https://imdb.com/title/tt1355642/?ref_=chttvtp_t_16   468 reviews
## 32  https://imdb.com/title/tt1355642/?ref_=chttvtp_t_16   468 reviews
## 33  https://imdb.com/title/tt2861424/?ref_=chttvtp_t_17   910 reviews
## 34  https://imdb.com/title/tt2861424/?ref_=chttvtp_t_17   910 reviews
## 35  https://imdb.com/title/tt1533395/?ref_=chttvtp_t_18    12 reviews
## 36  https://imdb.com/title/tt1533395/?ref_=chttvtp_t_18    12 reviews
## 37  https://imdb.com/title/tt8420184/?ref_=chttvtp_t_19   542 reviews
## 38  https://imdb.com/title/tt8420184/?ref_=chttvtp_t_19   542 reviews
## 39  https://imdb.com/title/tt0052520/?ref_=chttvtp_t_20   214 reviews
## 40  https://imdb.com/title/tt0052520/?ref_=chttvtp_t_20   214 reviews
## 41  https://imdb.com/title/tt1877514/?ref_=chttvtp_t_21   175 reviews
## 42  https://imdb.com/title/tt1877514/?ref_=chttvtp_t_21   175 reviews
## 43  https://imdb.com/title/tt1475582/?ref_=chttvtp_t_22 1,098 reviews
## 44  https://imdb.com/title/tt1475582/?ref_=chttvtp_t_22 1,098 reviews
## 45  https://imdb.com/title/tt2560140/?ref_=chttvtp_t_23 2,370 reviews
## 46  https://imdb.com/title/tt2560140/?ref_=chttvtp_t_23 2,370 reviews
## 47  https://imdb.com/title/tt0103359/?ref_=chttvtp_t_24   219 reviews
## 48  https://imdb.com/title/tt0103359/?ref_=chttvtp_t_24   219 reviews
## 49 https://imdb.com/title/tt11126994/?ref_=chttvtp_t_25 2,217 reviews
## 50 https://imdb.com/title/tt11126994/?ref_=chttvtp_t_25 2,217 reviews
##    Critic_Reviews Popularity_Rating
## 1             175                20
## 2             175                20
## 3               6               999
## 4               6               999
## 5              10             1,778
```

```
## 6              10             1,778
## 7              34               153
## 8              34               153
## 9              88               146
## 10             88               146
## 11             77               104
## 12             77               104
## 13             57               357
## 14             57               357
## 15              9             4,122
## 16              9             4,122
## 17             93                31
## 18             93                31
## 19             12             1,571
## 20             12             1,571
## 21              8             3,645
## 22              8             3,645
## 23             15             2,401
## 24             15             2,401
## 25            368                16
## 26            368                16
## 27              4               373
## 28              4               373
## 29              5             2,532
## 30              5             2,532
## 31             16               474
## 32             16               474
## 33             94               125
## 34             94               125
## 35              9             3,057
## 36              9             3,057
## 37             28             1,403
## 38             28             1,403
## 39             85               337
## 40             85               337
## 41             13             1,739
## 42             13             1,739
## 43            121               160
## 44            121               160
## 45             64                50
## 46             64                50
## 47             25               463
## 48             25               463
## 49             59                 1
## 50             59                 1
```

```r
#knitr::kable()

library(kableExtra)
```

```
##
## Attaching package: 'kableExtra'

## The following object is masked from 'package:dplyr':
##
```

```
##     group_rows
```

```
knitr::kable(shows,caption = "Extracting Rating, VoteCount, Episodes, Year and Reviews") %>%
kable_classic(full_width = T, html_font = "Cambria") %>%
kable_styling(font_size = 8)
```

Table 1: Extracting Rating, VoteCount, Episodes, Year and Reviews

Title	Rating	Voters	Episodes	Year	Show_Link	User_Reviews	Critic_Reviews	Popularity	Rating
Breaking Bad	9.5	2.2M	62	2008–2013	https://imdb.com/title/tt0903747/?ref_=chtvtp_t_1	5,108 reviews	137	20	1
Planet Earth II	9.5	163K	6	2016	https://imdb.com/title/tt0903747/?ref_=chtvtp_t_1	5,108 reviews	137	20	1
Planet Earth	9.4	224K	11	2006	https://imdb.com/title/tt5491994/?ref_=chtvtp_t_2	158 reviews	19	999	2
Band of Brothers	9.4	547K	10	2001	https://imdb.com/title/tt5491994/?ref_=chtvtp_t_2	158 reviews	19	999	2
Chernobyl	9.3	910K	5	2019	https://imdb.com/title/tt0795176/?ref_=chtvtp_t_3	111 reviews	15	1,778	3
The Wire	9.3	392K	60	2002–2008	https://imdb.com/title/tt0795176/?ref_=chtvtp_t_3	111 reviews	15	1,778	3
Avatar: The Last Airbender	9.3	391K	62	2005–2008	https://imdb.com/title/tt0389790/?ref_=chtvtp_t_4	1,058 reviews	15	153	4
Blue Planet II	9.3	49K	7	2017	https://imdb.com/title/tt0389790/?ref_=chtvtp_t_4	1,058 reviews	15	153	4
The Sopranos	9.2	500K	86	1999–2007	https://imdb.com/title/tt7366338/?ref_=chtvtp_t_5	3,534 reviews	14	146	5
Cosmos: A Spacetime Odyssey	9.2	132K	13	2014	https://imdb.com/title/tt7366338/?ref_=chtvtp_t_5	3,534 reviews	14	146	5
Cosmos	9.3	46K	13	1980	https://imdb.com/title/tt0307414/?ref_=chtvtp_t_6	787 reviews	10	104	6
Our Planet	9.2	54K	12	2019–2023	https://imdb.com/title/tt0307414/?ref_=chtvtp_t_6	787 reviews	10	104	6
Game of Thrones	9.2	2.4M	74	2011–2019	https://imdb.com/title/tt0457299/?ref_=chtvtp_t_7	1,003 reviews	35	357	7
Bluey	9.3	34K	194	2018–	https://imdb.com/title/tt0457299/?ref_=chtvtp_t_7	1,003 reviews	35	357	7
The World at War	9.2	32K	26	1973–1974	https://imdb.com/title/tt6769208/?ref_=chtvtp_t_8	53 reviews	1	122	8
Fullmetal Alchemist: Brotherhood	9.1	210K	68	2009–2010	https://imdb.com/title/tt6769208/?ref_=chtvtp_t_8	53 reviews	1	122	8
Rick and Morty	9.1	629K	78	2013–	https://imdb.com/title/tt0149842/?ref_=chtvtp_t_9	966 reviews	3	31	9
Life	9.1	44K	11	2009	https://imdb.com/title/tt0149842/?ref_=chtvtp_t_9	966 reviews	3	31	9
The Last Dance	9.0	160K	10	2020	https://imdb.com/title/tt2392695/?ref_=chtvtp_t_10	205 reviews	1	571	10
The Twilight Zone	9.0	97K	156	1959–1964	https://imdb.com/title/tt2392695/?ref_=chtvtp_t_10	205 reviews	1	571	10
The Vietnam War	9.1	30K	10	2017	https://imdb.com/title/tt0081846/?ref_=chtvtp_t_11	80 reviews	3	645	11
Sherlock	9.1	1M	15	2010–2017	https://imdb.com/title/tt0081846/?ref_=chtvtp_t_11	80 reviews	3	645	11
Attack on Titan	9.1	565K	98	2013–2023	https://imdb.com/title/tt9253866/?ref_=chtvtp_t_12	245 reviews	2	401	12
Batman: The Animated Series	9.0	123K	85	1992–1995	https://imdb.com/title/tt9253866/?ref_=chtvtp_t_12	245 reviews	2	401	12
Arcane	9.0	332K	18	2021–2024	https://imdb.com/title/tt9448847/?ref_=chtvtp_t_13	5,907 reviews	16	1	13
Breaking Bad	9.5	2.2M	62	2008–2013	https://imdb.com/title/tt9448847/?ref_=chtvtp_t_13	5,907 reviews	16	1	13

Planet Earth II	9.5	163K	6	2016	https://imdb.com/title/tt7678620/?ref_=chttvtp_t_14	369 reviews	373
Planet Earth	9.4	224K	11	2006	https://imdb.com/title/tt7678620/?ref_=chttvtp_t_14	369 reviews	373
Band of Brothers	9.4	547K	10	2001	https://imdb.com/title/tt0751075/?ref_=chttvtp_t_15	126 reviews	2,532
Chernobyl	9.3	910K	5	2019	https://imdb.com/title/tt0751075/?ref_=chttvtp_t_15	126 reviews	2,532
The Wire	9.3	392K	60	2002–2008	https://imdb.com/title/tt1355642/?ref_=chttvtp_t_16	468 reviews	474
Avatar: The Last Airbender	9.3	391K	62	2005–2008	https://imdb.com/title/tt1355642/?ref_=chttvtp_t_16	468 reviews	474
Blue Planet II	9.3	49K	7	2017	https://imdb.com/title/tt2861424/?ref_=chttvtp_t_17	910 reviews	125
The Sopranos	9.2	500K	86	1999–2007	https://imdb.com/title/tt2861424/?ref_=chttvtp_t_17	910 reviews	125
Cosmos: A Spacetime Odyssey	9.2	132K	13	2014	https://imdb.com/title/tt1533395/?ref_=chttvtp_t_18	12 reviews	3,057
Cosmos	9.3	46K	13	1980	https://imdb.com/title/tt1533395/?ref_=chttvtp_t_18	12 reviews	3,057
Our Planet	9.2	54K	12	2019–2023	https://imdb.com/title/tt8428184/?ref_=chttvtp_t_19	542 reviews	1,403
Game of Thrones	9.2	2.4M	74	2011–2019	https://imdb.com/title/tt8428184/?ref_=chttvtp_t_19	542 reviews	1,403
Bluey	9.3	34K	194	2018–	https://imdb.com/title/tt0052520/?ref_=chttvtp_t_20	214 reviews	337
The World at War	9.2	32K	26	1973–1974	https://imdb.com/title/tt0052520/?ref_=chttvtp_t_20	214 reviews	337
Fullmetal Alchemist: Brotherhood	9.1	210K	68	2009–2010	https://imdb.com/title/tt1877514/?ref_=chttvtp_t_21	175 reviews	1,739
Rick and Morty	9.1	629K	78	2013–	https://imdb.com/title/tt1877514/?ref_=chttvtp_t_21	175 reviews	1,739
Life	9.1	44K	11	2009	https://imdb.com/title/tt1475582/?ref_=chttvtp_t_22	1,098 reviews	160
The Last Dance	9.0	160K	10	2020	https://imdb.com/title/tt1475582/?ref_=chttvtp_t_22	1,098 reviews	160
The Twilight Zone	9.0	97K	156	1959–1964	https://imdb.com/title/tt2560140/?ref_=chttvtp_t_23	2,370 reviews	50
The Vietnam War	9.1	30K	10	2017	https://imdb.com/title/tt2560140/?ref_=chttvtp_t_23	2,370 reviews	50
Sherlock	9.1	1M	15	2010–2017	https://imdb.com/title/tt0123359/?ref_=chttvtp_t_24	219 reviews	463
Attack on Titan	9.1	565K	98	2013–2023	https://imdb.com/title/tt0123359/?ref_=chttvtp_t_24	219 reviews	463
Batman: The Animated Series	9.0	123K	85	1992–1995	https://imdb.com/title/tt1126994/?ref_=chttvtp_t_25	2,217 reviews	1
Arcane	9.0	332K	18	2021–2024	https://imdb.com/title/tt1126994/?ref_=chttvtp_t_25	2,217 reviews	1

```r
library(kableExtra)

movies <- shows[c(1:5),]

knitr::kable(movies, caption = "IMDB Movies") %>%
kable_classic(full_width = T, html_font = "Arial Narrow") %>%
kable_styling(font_size = 8)
```

Table 2: IMDB Movies

Title	Rating	Voters	Episodes	Year	Show_Link	User_Reviews	Critic_Reviews	Popularity_Rating
Breaking Bad	9.5	2.2M	62	2008–2013	https://imdb.com/title/tt0903747/?ref_=chtvtp_t_1	5.1K reviews	37	20
Planet Earth II	9.5	163K	6	2016	https://imdb.com/title/tt0903747/?ref_=chtvtp_t_1	5.1K reviews	37	20
Planet Earth	9.4	224K	11	2006	https://imdb.com/title/tt5491994/?ref_=chtvtp_t_2	158 reviews	61	999
Band of Brothers	9.4	547K	10	2001	https://imdb.com/title/tt5491994/?ref_=chtvtp_t_2	158 reviews	61	999
Chernobyl	9.3	910K	5	2019	https://imdb.com/title/tt7575176/?ref_=chtvtp_t_3	111 reviews	15	1,778

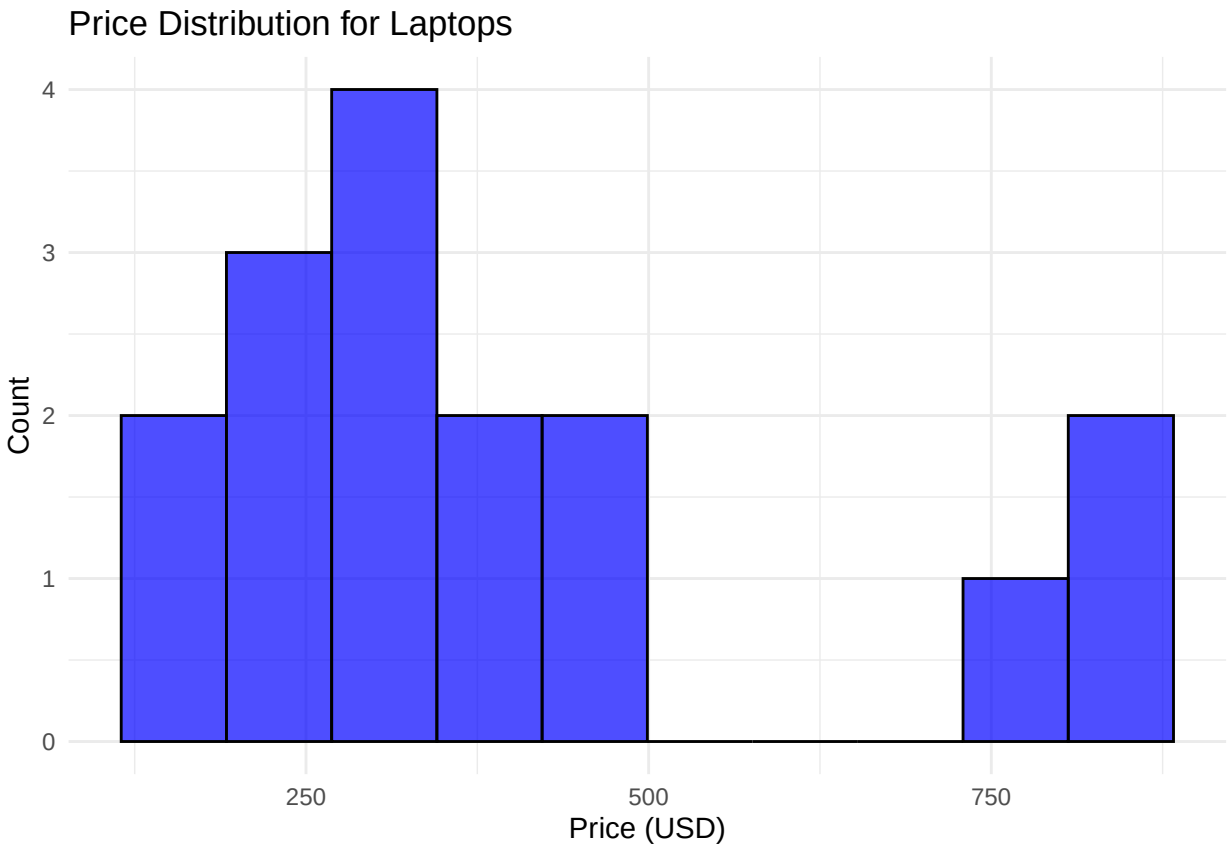
Extracting Amazon Product Reviews

```
# Define the scraping function
scrape_amazon <- function(url) {
  page <- read_html(url)

  # Extract product details
  products <- page %>% html_nodes(".s-title-instructions-style") %>% html_text(trim = TRUE)
  prices <- page %>% html_nodes(".a-price-whole") %>% html_text(trim = TRUE)
  ratings <- page %>% html_nodes(".a-icon-alt") %>% html_text(trim = TRUE)
  reviews <- page %>% html_nodes(".s-underline-text") %>% html_text(trim = TRUE)

  # Handle missing data by aligning lengths
  max_length <- max(length(products), length(prices), length(ratings), length(reviews))
  products <- c(products, rep(NA, max_length - length(products)))
  prices <- c(prices, rep(NA, max_length - length(prices)))
  ratings <- c(ratings, rep(NA, max_length - length(ratings)))
  reviews <- c(reviews, rep(NA, max_length - length(reviews)))

  # Create a data frame
  return(data.frame(
    Product = products,
    Price = as.numeric(gsub("[^0-9.]", "", prices)),
    Ratings = as.numeric(gsub("[^0-9.]", "", str_extract(ratings, "^[0-9.]+"))),
    Reviews = as.numeric(gsub("[^0-9]", "", reviews)),
    stringsAsFactors = FALSE
  ))
}
```

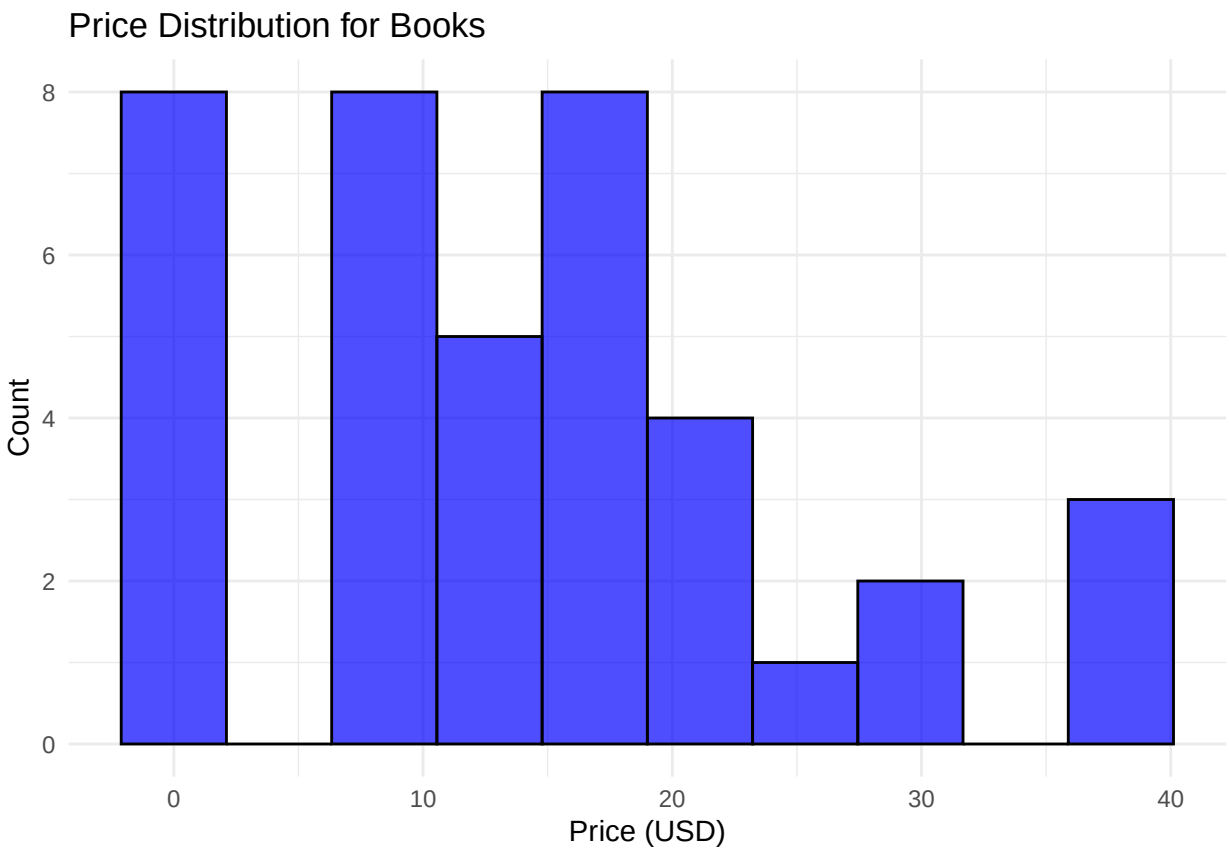
```
# Define URLs for categories
categories <- c("Laptops", "Books", "Shoes", "Televisions", "Fashion Bags")
urls <- c(
  'https://www.amazon.com/s?k=laptop&crid=108GXR4VZZEMS&sprefix=lap%2Caps%2C680&ref=nb_sb_ss_ts-doa-p_1_
  'https://www.amazon.com/s?k=books&i=stripbooks-intl-ship&crid=3C5FBQTXKB575&sprefix=books%2Cstripbooks
  'https://www.amazon.com/s?k=shoes&i=stripbooks-intl-ship&crid=PWE5DZDD7EU7&sprefix=shoes%2Cstripbooks-
  'https://www.amazon.com/s?k=television&i=stripbooks-intl-ship&crid=08JO99JDMGHY&sprefix=television%2Cs
  'https://www.amazon.com/s?k=fashion+bags&i=stripbooks-intl-ship&crid=3N3PC8YMHSW66&sprefix=fashion+bag
)

# Scrape data for all categories
amazon_data <- lapply(urls, scrape_amazon)
names(amazon_data) <- categories

# Combine all data into a single data frame
combined_data <- bind_rows(amazon_data, .id = "Category")
```

Plot price distributions

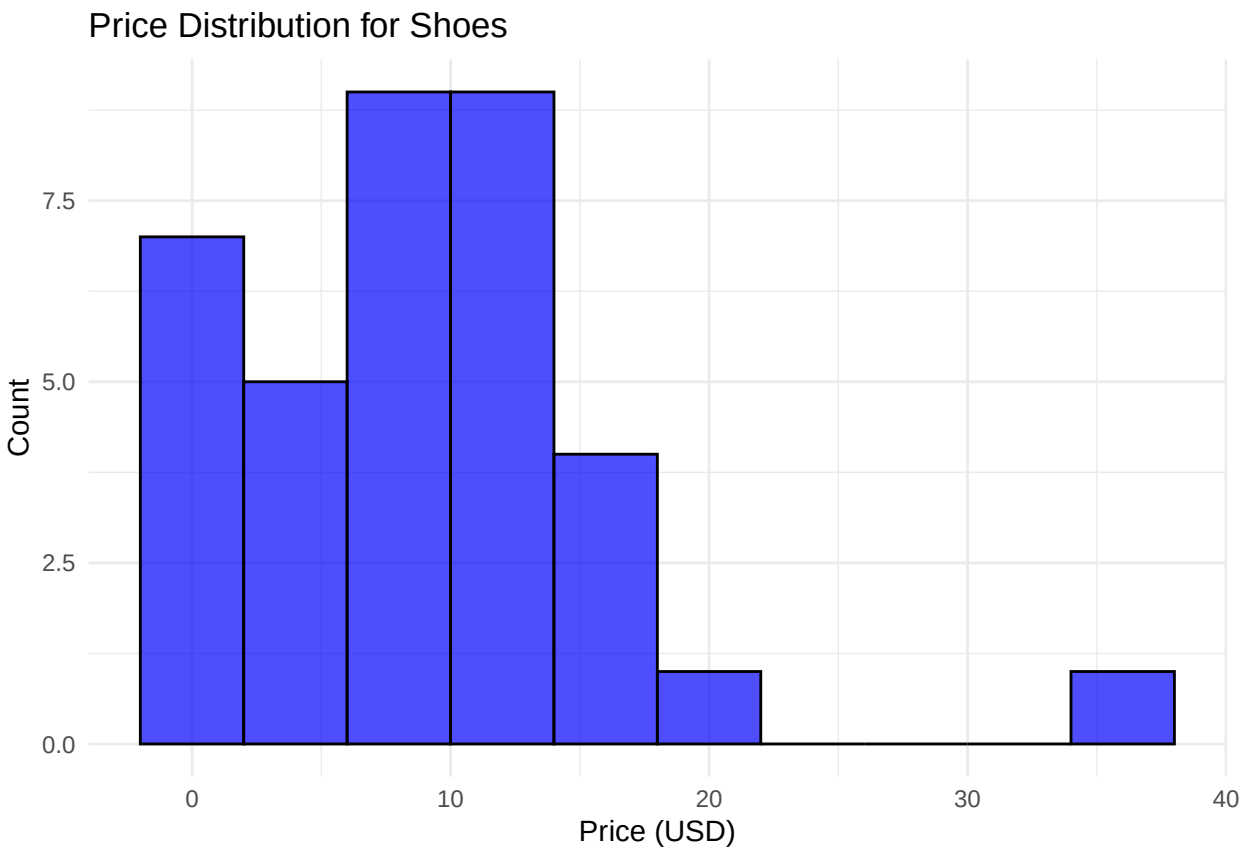
```
## Warning: Removed 78 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



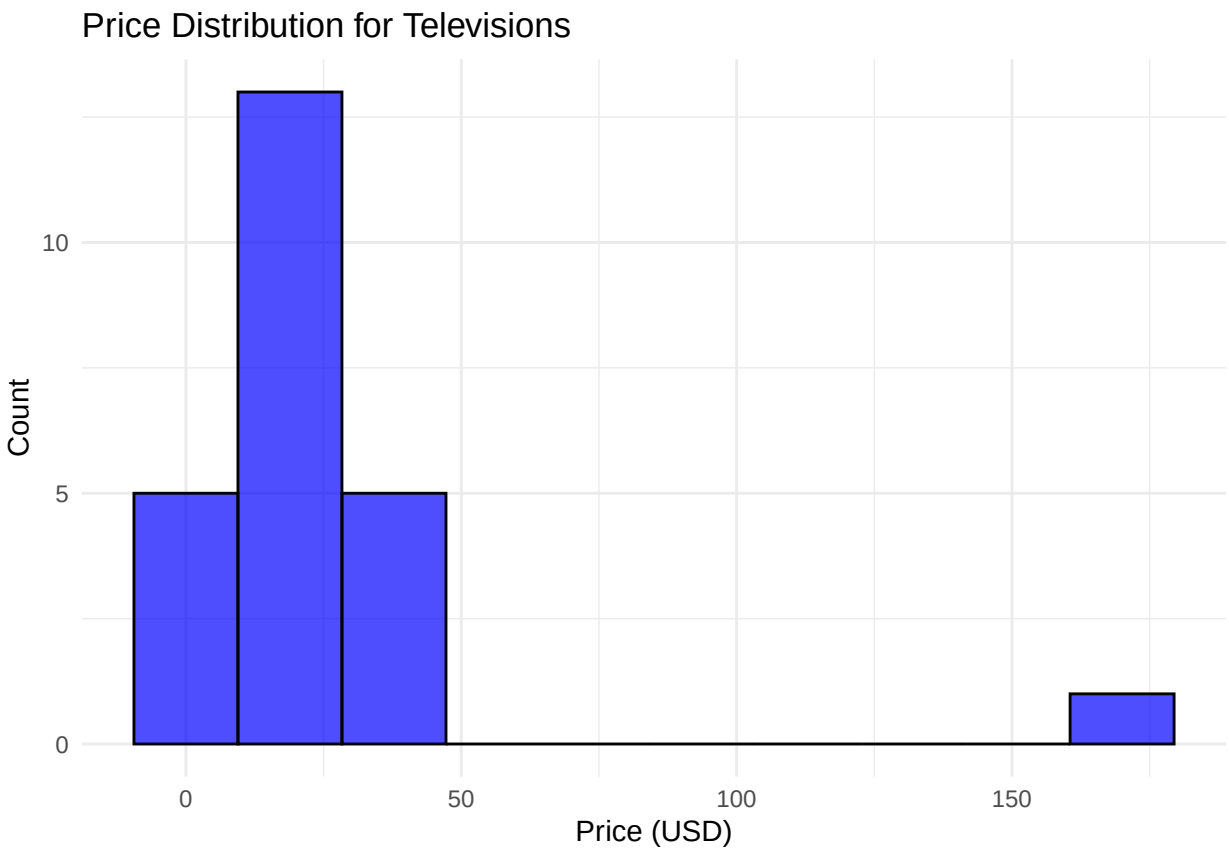

```
## Warning: Removed 125 rows containing non-finite outside the scale range
## (`stat_bin()`).
```



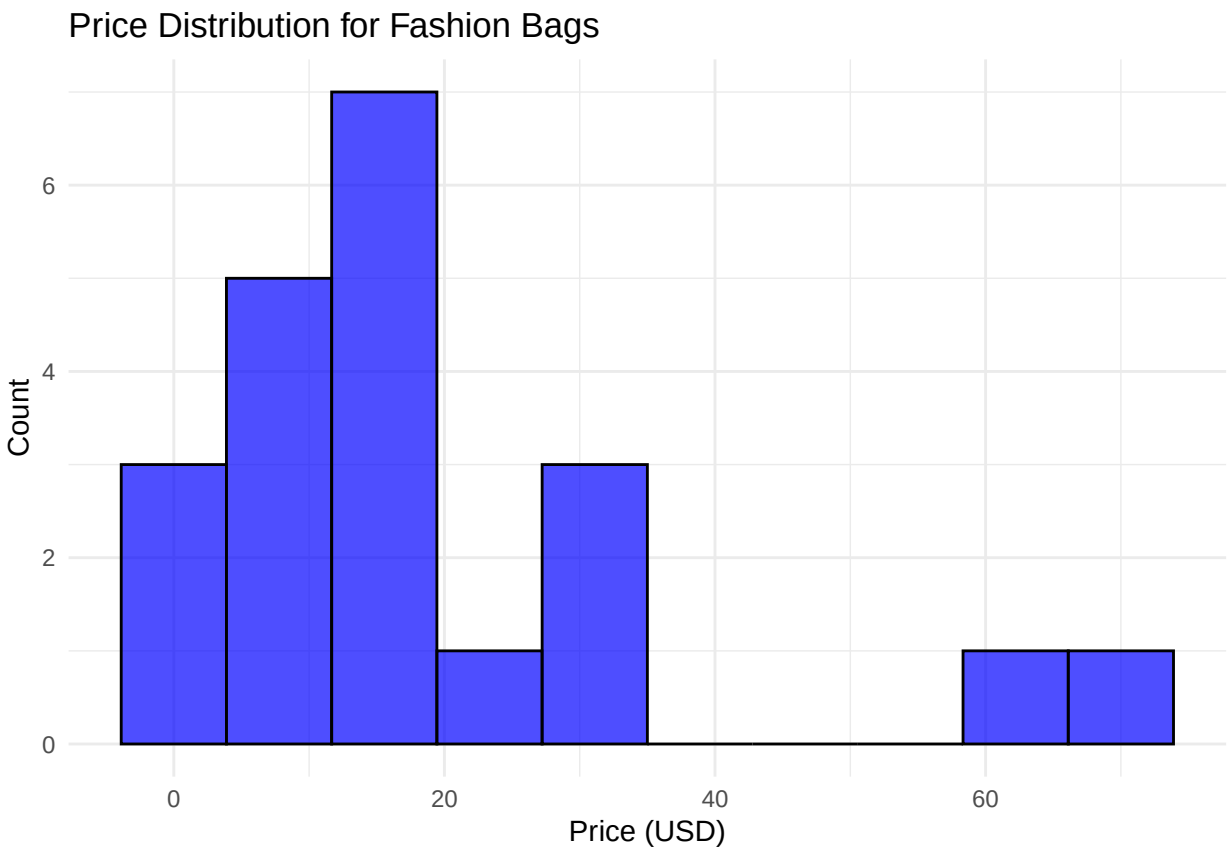

```
## Warning: Removed 135 rows containing non-finite outside the scale range
## (`stat_bin()`).
```




```
## Warning: Removed 89 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

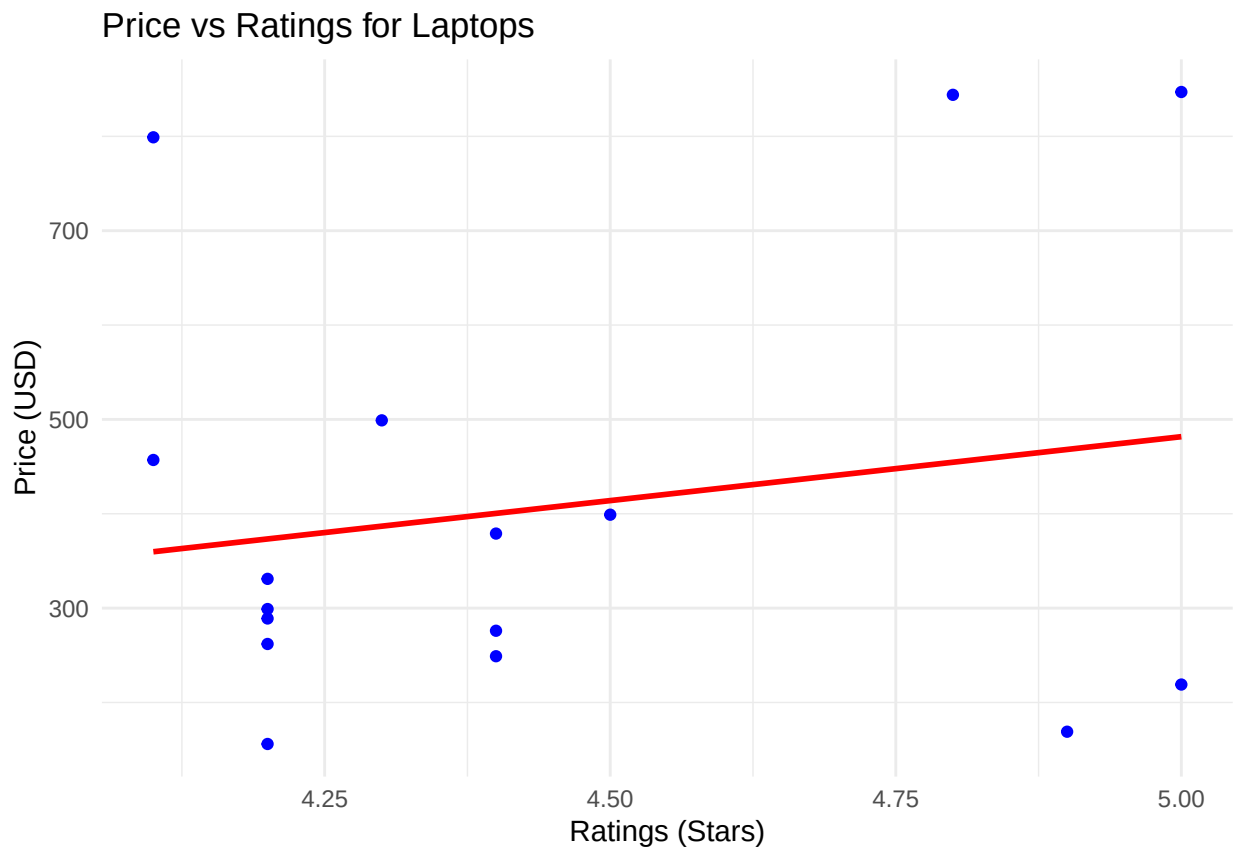



```
## Warning: Removed 100 rows containing non-finite outside the scale range
## (`stat_bin()`).
```

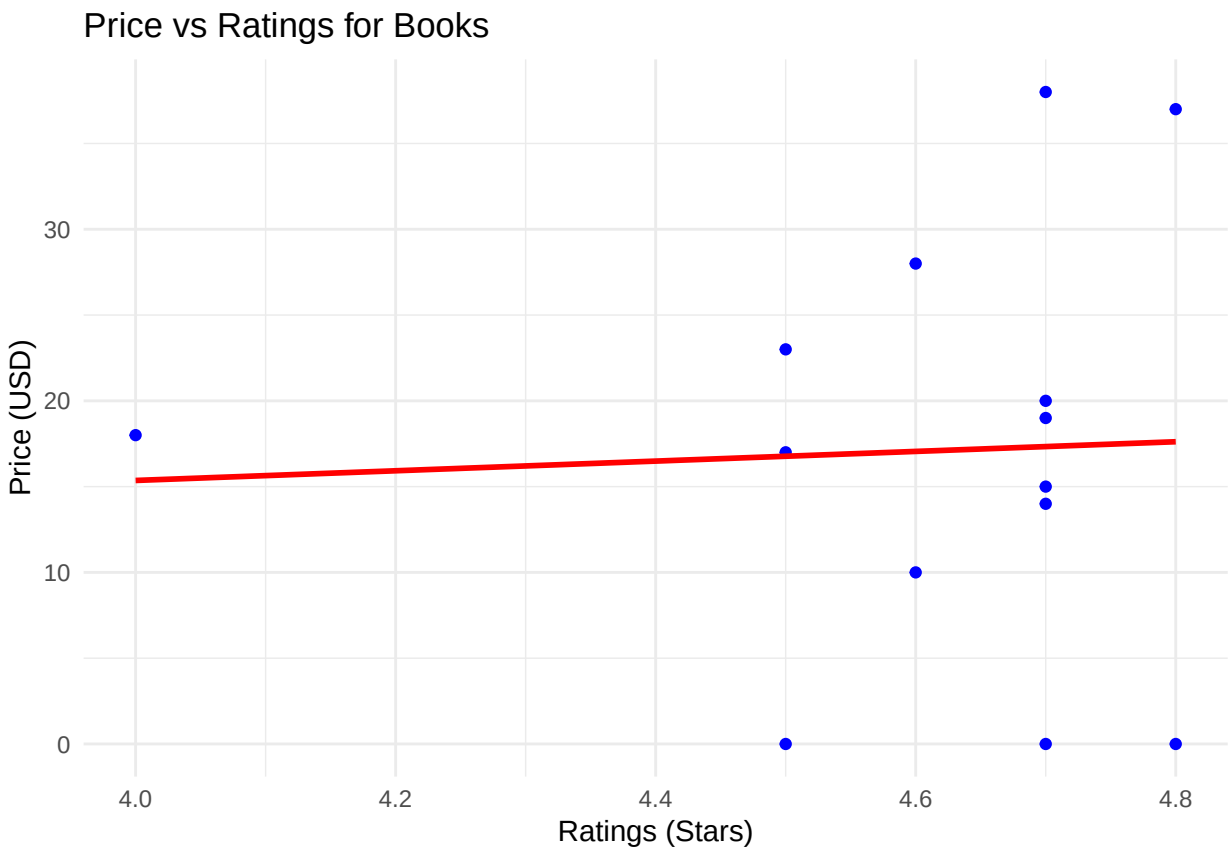


Plot price vs ratings

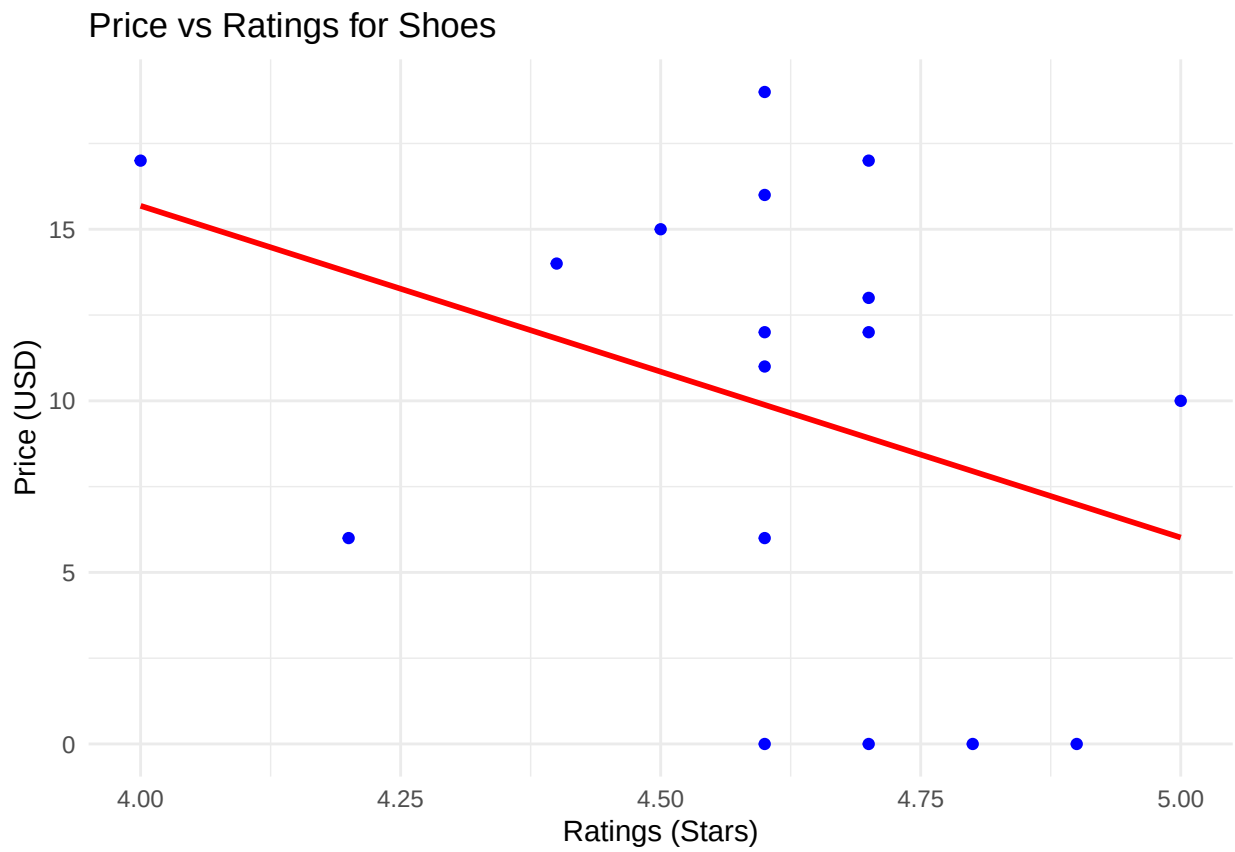
```
## `geom_smooth()` using formula = 'y ~ x'
```



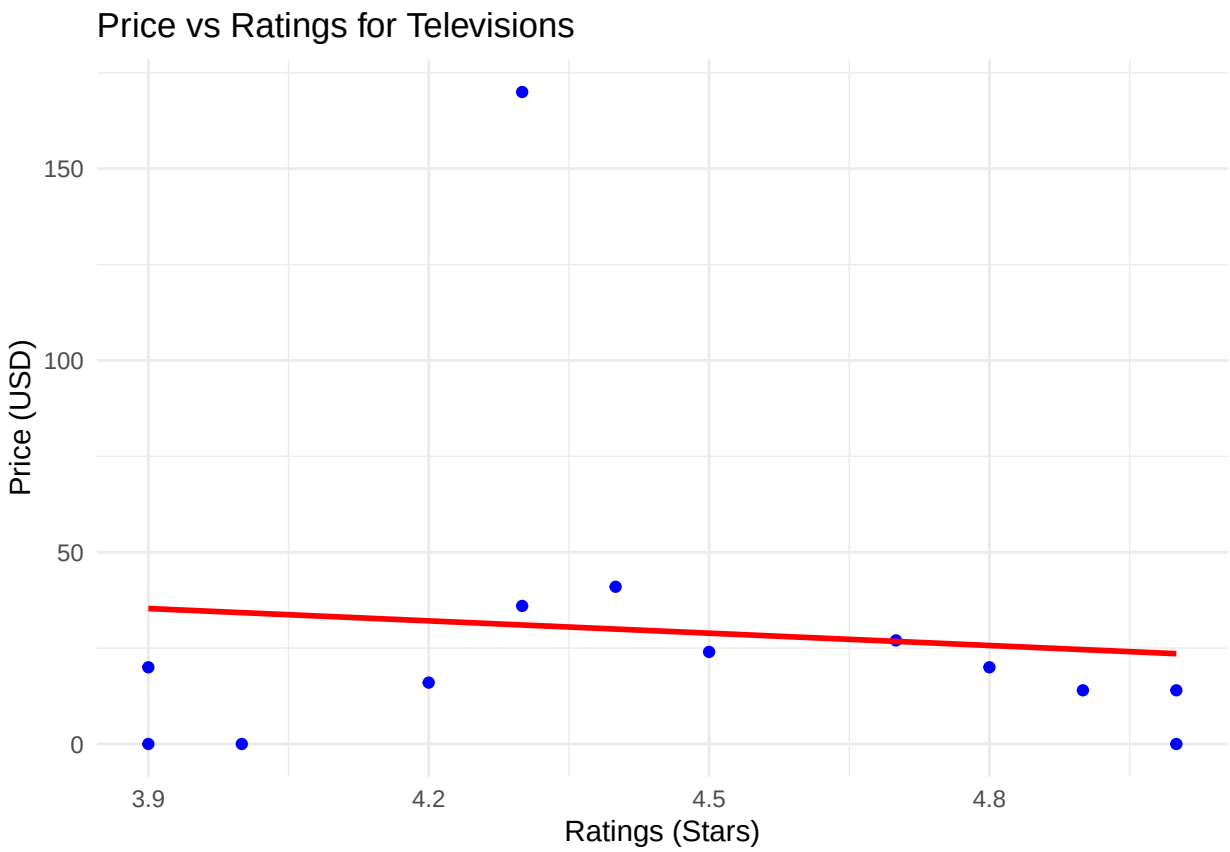

```
## `geom_smooth()` using formula = 'y ~ x'
```



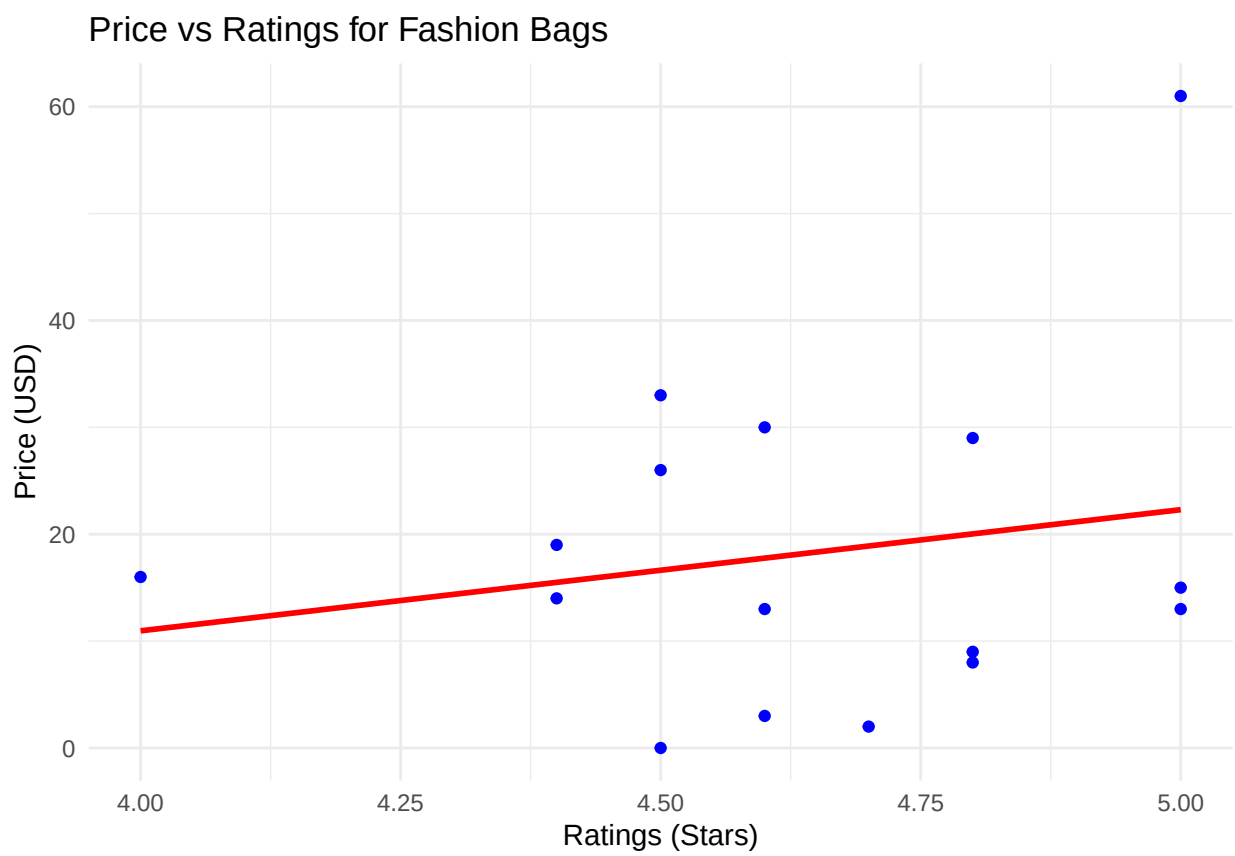

```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
## `geom_smooth()` using formula = 'y ~ x'
```




```
# Rank products within each category
rank_products <- function(data) {
  data <- data %>%
    arrange(desc(Ratings), Price) %>%
    mutate(Rank = row_number())
  return(data)
}

ranked_data <- lapply(amazon_data, rank_products)

# Print top 5 products per category
for (category in categories) {
  cat("\nTop 5 Products in", category, "\n")
  print(head(ranked_data[[category]], 5))
}
```

```
##
## Top 5 Products in Laptops
##
## 1                                                                    ASUS 15.6" Vivobook Go Lap
## 2                              HP 17 Laptop, 17.3" HD+ Touchscreen Display, 12th Gen Intel Core i7-1
## 3                                     ASUS Chromebook CX1, 15.6" FHD NanoEdge Display, Intel Celer
## 4 Apple 2024 MacBook Air 13-inch Laptop with M3 chip: Built for Apple Intelligence, 13.6-inch Liquid
## 5    HP Newest 255 G10 Laptop for Home or Work, 16GB RAM, 1TB SSD, 15.6" Full HD, Ryzen 3 7330U (Bea
##   Price Ratings    Reviews Rank
## 1   219     5.0 2.0070e+03    1
## 2   847     5.0         NA    2
```

```
## 3   169     4.9 1.5645e+17    3
## 4   844     4.8 6.0000e+00    4
## 5   399     4.5 6.0000e+00    5
##
## Top 5 Products in Books
##
## 1               Good Energy: The Surprising Connection Between Metabolism and Limitless Health  by Ca
## 2          Wind and Truth: Book Five of the Stormlight Archive  Book 5 of 5: The Stormlight Archive
## 3 Hot Mess (Diary of a Wimpy Kid Book 19) (Diary of a Wimpy Kid, 19) (Volume 19)  Part of: Diary of a
## 4                                        Forgotten Home Apothecary : 250 Powerful Remedies at Your Fi
## 5                           Onyx Storm (Deluxe Limited Edition) (The Empyrean, 3)  Book 3 of 3: The E
##   Price Ratings     Reviews Rank
## 1     0     4.8          NA    1
## 2    37     4.8 1.59916e+15    2
## 3     0     4.7          NA    3
## 4    14     4.7 2.70027e+07    4
## 5    15     4.7 1.66000e+02    5
##
## Top 5 Products in Shoes
##
## 1                                    365 Days of Shoes Picture-A-Day® Wall Calendar 2025: An Obsessiv
## 2                               Pete the Cat: I Love My White Shoes  Part of: Pete the Cat (63 books)
## 3                                                                                   Those Shoes  B
## 4 Sensible Shoes: A Story about the Spiritual Journey (Sensible Shoes Series)  Book 1 of 4: Sensible
## 5                                             Dancing Shoes (The Shoe Books)  Part of: The Shoe Book
##   Price Ratings Reviews Rank
## 1    10     5.0      29    1
## 2     0     4.9      NA    2
## 3     0     4.8      NA    3
## 4     0     4.7      NA    4
## 5    12     4.7 3652025    5
##
## Top 5 Products in Televisions
##
## 1                     3... 2...1... We're on the Air: An Inside Look at Sports Television, Journalism
## 2                                                                 The Big Story: The Oral History of E
## 3                             Black TV: Five Decades of Groundbreaking Television from Soul Train to
## 4 Behind the Screens: Illustrated Floor Plans and Scenes from the Best TV Shows of All Time  by Iñaki
## 5                 Goodnight John-Boy: A Memory Book of The Waltons, One of Television's Greatest Fami
##   Price Ratings     Reviews Rank
## 1     0     5.0 1.49915e+15    1
## 2    14     5.0          NA    2
## 3    14     4.9 7.14000e+02    3
## 4    20     4.8          NA    4
## 5    27     4.7          NA    5
##
## Top 5 Products in Fashion Bags
##
## 1 Designing Bags: Typology, Construction Techniques, Analogue and Digital Patternmaking from Scratch
## 2                                                                                Fashion Sketchbook
## 3                                     The Handbag Book: 400 Designer Bags That Changed Fashion  by Se
## 4                                          Yves Saint Laurent: Gold: Fashion, Jewelry, Shoes, and Ba
## 5                                  Aunt Jo's Scrap-Bag Volume VI: An Old-Fashioned Thanksgiving, and
##   Price Ratings      Reviews Rank
```

```
## 1    13     5.0           NA    1
## 2    15     5.0 2.100000e+01    2
## 3    61     5.0 4.000000e+02    3
## 4     8     4.8 6.197620e+15    4
## 5     9     4.8 3.993994e+13    5
```